untitled (3.14.2)

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```
#Sampling
#Program 1: Populations, Samples and Random Sampling
#Aim: To understand the difference between a population and a sample, and to draw a simple random sample using Python.
#Scenario: A university has 10,000 students.
#Their exam marks (out of 100) form the population.
#The examination controller cannot check all records, so she selects 50 students at random (a sample) to estimate the average mark.

import numpy as np
import matplotlib.pyplot as plt
np.random.seed(42) # to get the same result every time
# Step 1: Create the population - marks of 10,000 students
population = np.random.normal(loc=65, scale=12, size=10000).round()
population = np.clip(population, 0, 100) # marks must be between 0 and 100
# Step 2: Draw a simple random sample of 50 students (without replacement)
sample = np.random.choice(population, size=50, replace=False)
# Step 3: Compare population parameters and sample statistics
print("POPULATION (N = 10000)")
print(f" Population mean (mu) = {population.mean():.2f}")
print(f" Population SD (sigma) = {population.std():.2f}")
print("SAMPLE (n = 50)")
print(f" Sample mean (x-bar) = {sample.mean():.2f}")
print(f" Sample SD (s) = {sample.std(ddof=1):.2f}")
print(f"Sampling error (x-bar - mu) = {sample.mean() - population.mean():.2f}")
# Step 4: Different random samples give different sample means
print("\nMeans of 5 different random samples of size 50:")
for i in range(1, 6):
    s = np.random.choice(population, size=50, replace=False)
    print(f" Sample {i}: mean = {s.mean():.2f}")
# Step 5: Graph
fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].hist(population, bins=30, color='skyblue', edgecolor='black')
ax[0].axvline(population.mean(), color='red', linestyle='--', label=f'Population mean = {population.mean():.1f}')
ax[0].set_title('Population: Marks of 10,000 Students')
ax[0].set_xlabel('Marks');
ax[0].set_ylabel('Number of students');
ax[0].legend()
ax[1].hist(sample, bins=10, color='orange', edgecolor='black')
ax[1].axvline(sample.mean(), color='green', linestyle='--', label=f'Sample mean = {sample.mean():.1f}')
ax[1].axvline(population.mean(), color='red', linestyle=':', label=f'Population mean = {population.mean():.1f}')
ax[1].set_title('Random Sample of 50 Students')
ax[1].set_xlabel('Marks');
ax[1].set_ylabel('Number of students');
ax[1].legend()
plt.tight_layout()
plt.show()
```

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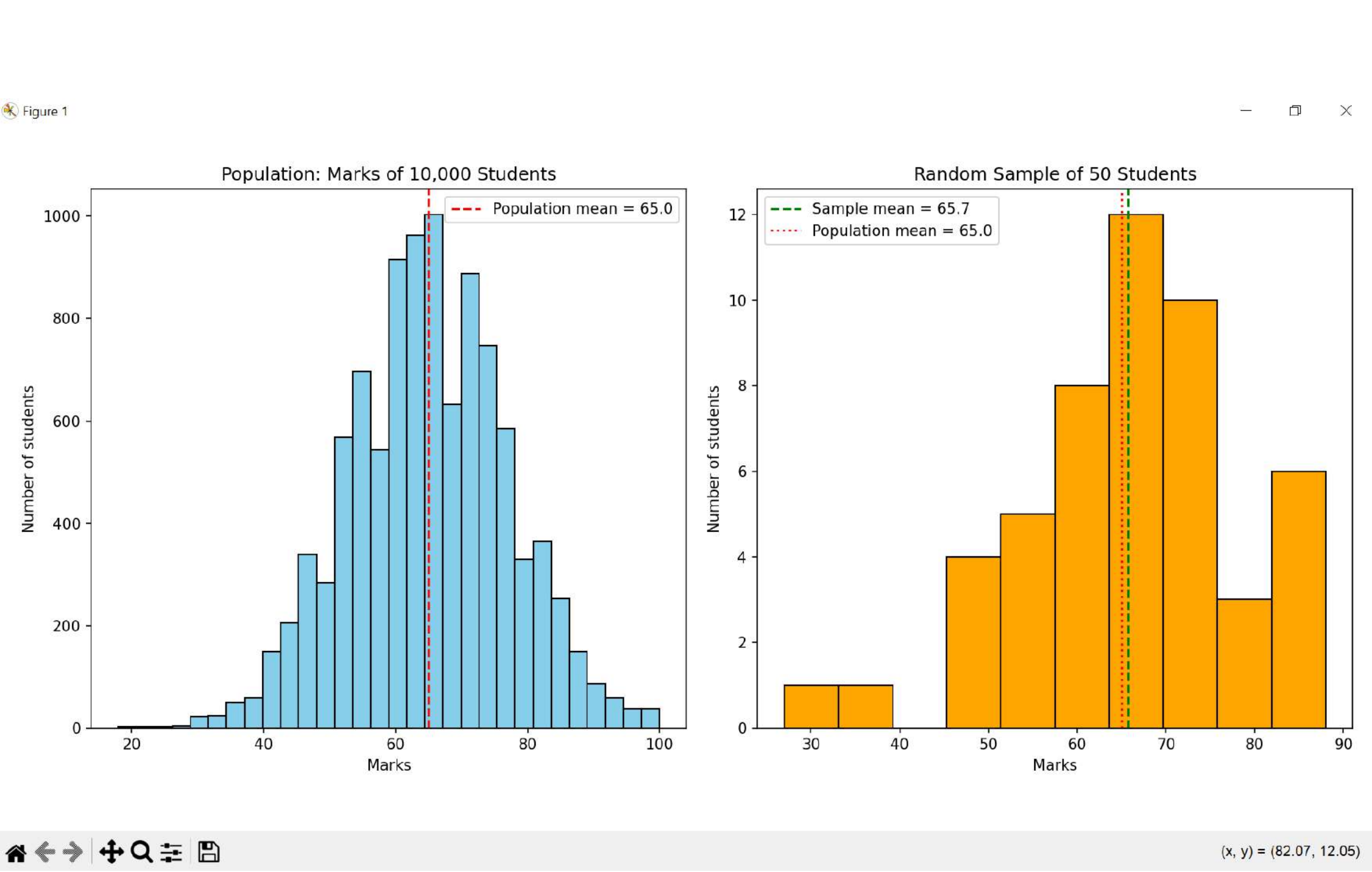
IDLE Shell 3.14.2

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Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/pings/Desktop/son.pu.py ==========
POPULATION (N = 10000)
 Population mean (mu) = 64.96
 Population SD (sigma) = 12.02
SAMPLE (n = 50)
 Sample mean (x-bar) = 65.70
 Sample SD (s) = 12.67
Sampling error (x-bar - mu) = 0.74

Means of 5 different random samples of size 50:
 Sample 1: mean = 64.14
 Sample 2: mean = 65.54
 Sample 3: mean = 64.92
 Sample 4: mean = 65.02
 Sample 5: mean = 64.74
>>>
```

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Figure 1
Population: Marks of 10,000 Students
Population mean = 65.0
Number of students
Marks
0
200
400
600
800
1000
20
40
60
80
100
Random Sample of 50 Students
Sample mean = 65.7
Population mean = 65.0
Number of students
Marks
0
2
4
6
8
10
12
30
40
50
60
70
80
90
(x, y) = (82.07, 12.05)